ENERGYMIND TUNISIA

WattWitness

Microgrid Data Trust Layer

Formal Project Proposal · Phase 1: Ideation & Data Exploration

Chosen system slice: one simulated microgrid node
PV + wind + variable synthetic building load + battery + grid connection

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Proposal principle

Build and demonstrate a transparent, human-supervised trust layer between simulated energy telemetry and any later automated decision. Report what is implemented, what remains a design, and what still requires validation.

1. Executive summary

WattWitness is a Phase 1 prototype for the EnergyMind Tunisia challenge. It treats every telemetry reading as testimony and cross-examines it against the physical relationships it must satisfy and against simple data-quality rules. Each alert carries a plain-language reason and an evidence score and goes to a human reviewer, whose decision is recorded in an audit log. The goal is not to replace an operator: the system should state what its evidence supports and escalate uncertain cases.

The prototype uses a one-year hourly simulation-derived dataset. PV and wind values are retained from a supplied base simulation export; the building load and context labels are synthetic; grid/battery dispatch is recalculated under stated assumptions; and voltage/current/inverter channels are derived through simplified formulas. A meter model then adds realistic measurement error, finite resolution and telemetry outages, so the data are not artificially perfect. Controlled faults are injected into a held-out window to test detection. Those results are simulated test results, not evidence of real-world performance.

2. Scope

In scope for Phase 1	Out of scope / future work
One simulated microgrid telemetry node; hourly series for PV, wind, building load, grid import, battery and derived electrical signals.	Production deployment, real-time connection to utility infrastructure, autonomous equipment control, or automatic data repair.
Controlled injection of sensor/data-quality faults with traceable event metadata.	Claims that injected events replicate every real equipment failure or cyberattack.
Transparent rules and physical-consistency checks; comparisons against baselines.	A fully validated ML model or calibrated probability of fault.
Human review queue with recorded decisions and an escalation policy.	Multi-node exchange or federated learning implementation; Phase 1 includes design documentation only.

3. Objectives

- Detect selected classes of missing, abrupt, frozen, delayed, drifting and biased telemetry in a repeatable experiment.
- Compare suspicious readings with related power, voltage and current signals to surface inconsistencies.
- Give each alert a plain-language rationale, evidence strength, recommended check and human-review state.
- Define a human review step in which decisions and reasons are recorded and no equipment is controlled.
- Define clear escalation boundaries and propose a privacy-preserving mechanism for future sharing between nodes.
- Deliver the prepared dataset, source code, experiment results, technical report and documented limitations.

4. Data and anomaly-simulation plan

4.1 Baseline dataset

The prepared dataset contains 8,784 hourly rows from 1 June 2023 through 31 May 2024. The renewable PV/wind series are retained from the supplied simulation export. The original repeated daily load is used only as a rough shape template; a different synthetic building load is generated with weekday/weekend differences, seasonal multipliers, daily variability, correlated hourly noise and intermittent demand events. EV load is excluded.

Because derived channels satisfy their equations exactly, a seeded meter model converts this ideal state into measured telemetry: about 0.5% error on power and current, 0.2% on voltage, finite reporting resolution, and 18 natural outages leaving 39 rows with unknown values. Detection limits are learned from the training period of this measured data.

Signal family	Construction and interpretation
PV and wind power	Retained from the supplied base simulation series; units and measurement/simulation provenance are inherited and still need confirmation.
Building load and context	Synthetic variable building demand plus generated context labels. Not measured demand or activity.
Grid and battery	Recalculated using a stated balance equation, 100 kW battery limit, assumed 100–500 kWh stored-energy range and 95% charge/discharge efficiency.
Voltage, current and inverter	Derived with simplified formulae using assumed 400 V three-phase voltage and power factors. These are not real sensor measurements or detailed equipment simulations.

4.2 Injection protocol

The final 60 days of the prepared dataset are held out. A fixed seed creates 18 traceable events (three per type, from gross to subtle severity) altering 159 hourly readings; only readings whose value actually changed are labelled. Each injected event records its type, target signal, start/end times, duration, severity and seed. The untouched baseline is retained separately; injection labels are excluded from the detector inputs and are used only to evaluate predictions.

Injected scenario	Demonstration intent
Spike in grid power	Detect a one-sample extreme jump.
Frozen PV reading	Detect a meaningful signal remaining unchanged for too long.
Gradual building-load drift	Test a slow offset against the modelled cross-signal relationships.
Missing battery SOC	Detect missing telemetry values.
Delayed/replayed PV signal	Test a reading that is plausible in magnitude but misaligned in time.
Plausible grid-current bias	Test an altered measurement that can conflict with voltage/power consistency.

4.3 Evaluation

Compare simple rules, physical-consistency checks, their transparent combination, and an Isolation Forest baseline on the same held-out test window. Report precision, recall, F1 and average precision, event-level detection, recall by fault type, alerts on the untouched window, and sensitivity to meter error. The untouched-window figure is described as an alert rate, not a verified false-positive rate, because there are no independent labels for every real event in the base simulation.

5. Preliminary design for each official Challenge Goal

The four goal names below follow the official EnergyMind Tunisia challenge description. The implementation status is stated conservatively; a preliminary design is not the same as a completed implementation.

Goal 1 — Anomaly & Bias Detection

Design: monitor the available energy variables and derive checks for missing values, abrupt jumps, frozen values and disagreement between linked electrical channels. The hybrid alert is raised when a data-quality rule triggers or a physical-consistency residual exceeds the current threshold. The controlled experiment covers spike, frozen value, gradual drift, missing value, delayed/replayed signal and plausible current bias.

Distinguishing event types: a reading that conflicts with related channels supports a data-quality hypothesis. A coherent change across multiple channels without a strong physical mismatch is shown as a potential operating event requiring context confirmation. In Phase 1 the system cannot establish causality with certainty because the simulation lacks independent labels for genuine grid events and the synthetic context is not a trusted operational log. Power-quality waveforms and sub-second smart-meter streams are outside the included hourly dataset, so the present implementation focuses on the signals actually available.

Status: partially implemented. The single-node detector and controlled test are implemented. Context-aware discrimination between genuine events, natural equipment failures and telemetry corruption requires stronger context data and independently labelled cases.

Goal 2 — Confidence Scores & Explainability

Design: each alert carries a plain-language reason, the related residuals/rules, a qualitative evidence label, a suggested operator check, and an audit-row timestamp. Each alert also carries a simple heuristic evidence score and the size of every mismatch relative to its learned limit. The score is explicitly labelled as uncalibrated and is not presented as the probability that a fault occurred.

Status: partially implemented. Explanations and logging are generated. Confidence calibration, SHAP/LIME analysis, and independent human scoring of explanation quality are not yet completed.

Goal 3 — Human-AI Escalation & Governance

Design: the AI may flag and explain, but will not change raw readings or control equipment. A human reviewer can confirm a data issue, judge that a change may be a genuine operating event, dismiss an alert, or escalate it, while recording the reason. Decisions are appended to an audit log whose existing entries are never rewritten. To reduce over-trust, the score is labelled as heuristic and shows the evidence and limitations; to reduce under-trust, it makes the checks and supporting signals inspectable and gives the operator a documented override/escalation path.

Provisional evidence tier	Proposed response in Phase 1
No meaningful rule or physical conflict; heuristic score below 0.35	No automatic action. Continue monitoring; do not treat the lack of an alert as proof of safety.
Rule pattern without a strong physical mismatch; score around 0.35–0.69	Queue for operator review. Ask the operator to inspect adjacent values, timing and context.
Strong physical mismatch; score 0.70 or higher	Prioritize human review and compare related signals/source records. Do not auto-correct or trip equipment.
Missing critical signal, contradictory evidence or uncertain diagnosis	Escalate to an authorized operator; preserve the raw data and log the decision.

These score bands are illustrative governance thresholds for the demo, not calibrated values or a safety-certified policy. Automatic corrective actions remain disabled. Status: review workflow prototyped; production governance remains future work.

Goal 4 — Shared Learning Across the Noosphere

Design: nodes may share approved anonymized anomaly patterns, aggregated calibration insights and relevant knowledge as normalized summaries (signal family, coarse shape/duration, checks triggered, detector version and reviewed outcome), not raw traces, precise timestamps, device identifiers or private schedules by default. Each receiving node checks units, signals, sign conventions and physical compatibility; patterns run in shadow mode and require provenance and review before adoption. The design considers privacy leakage, poisoned knowledge, negative transfer and false confidence.

Status: design documented only. No multi-node exchange, shared model or federated training is implemented in Phase 1; the challenge states that multi-node sharing may remain a documented design.

6. System architecture and operator workflow

1. Prepare the one-year simulated dataset; leave the clean source table unchanged.
2. Apply the meter model to obtain measured telemetry with realistic error and gaps.
3. Create a labeled faulted copy of the held-out test window using fixed injection settings.
4. Run data-quality rules and simplified physical-consistency checks; compare against baselines.
5. Generate alert explanations, evidence labels, metrics and a review queue.
6. A reviewer inspects each alert with its related signals and records a decision and a reason.
7. Retain a local audit record of reviewer label, decision and notes. No equipment command is sent.

7. Implementation status and deliverables

Phase 1 element	Status
Prepared simulated dataset and data dictionary	Included; base export is not included in the public bundle. Confirm rights for derived PV/wind series before public upload.
Fault injection, detector and evaluation scripts	Implemented; repeatable with included prepared dataset.
Metric output and plots	Included; results are on controlled synthetic faults only.
Explainable alert output and review queue	Implemented; evidence score is heuristic, not calibrated.
Human review queue and decision log	Implemented as a prototype; not a secured production audit system.
Cross-node privacy/sharing approach	Documented design only, as permitted for Phase 1.
Formal proposal and technical report	Included as Word and PDF.

8. Main risks and next validation steps

- Verify inherited PV/wind provenance, units and redistribution rights for the prepared dataset.
- Verify the battery-state interpretation, dispatch assumptions and sign convention against the originating simulation model.
- Add examples of coherent genuine operating events and independently labelled sensor faults; evaluate event-versus-data classification.
- Calibrate evidence scores on labelled data and assess explanation quality with operators.
- If authorized, test replayed or non-invasive data with an independent laboratory node; do not connect the prototype to live control equipment.
- For multi-node learning, implement sender identity, versioned schema, provenance checks, privacy tests and shadow-mode evaluation before activation.